

```
n <- as.integer(
  readline("Enter number of records: ")
)
spend <- numeric(n)
sales <- numeric(n)
for(i in 1:n) {
  spend[i] <- as.numeric(
    readline(paste("Enter Spend", i, ": "))
  )
  sales[i] <- as.numeric(
    readline(paste("Enter Sales", i, ": "))
  )
}
advertising <- data.frame(
  Spend = spend,
  Sales = sales
)
cat("\nAdvertising Data:\n")
print(advertising)
```

```
model <- lm(
  Sales ~ Spend,
  data = advertising
)
cat("\nRegression Model:\n")
print(model)
```

```
# Prediction
```

```
new_spend <- as.numeric(
```

```
  readline("\nEnter Spend for prediction: ")
```

```
)
```

```
new_data <- data.frame(
```

```
  Spend = new_spend
```

```
)
```

```
prediction <- predict(
```

```
  model,
```

```
  newdata = new_data
```

```
)
```

```
cat("\nPredicted Sales =", prediction)
```

The screenshot displays the RStudio environment with the following components:

- Source Editor:** Contains R code for loading data, fitting a linear model, and making a prediction.
- Console:** Shows the execution of the code, including user input for sales and spend values, and the resulting predicted sales value.
- Environment Pane:** Lists the objects in the global environment: advertising, model, new_data, and new_spend.

R Code (Source Editor):

```
18 cat("\nAdvertising Data:\n")
19 print(advertising)
20
21 model <- lm(
22   Sales ~ Spend,
23   data = advertising
24 )
25 cat("\nRegression Model:\n")
26 print(model)
27 # Prediction
28 new_spend <- as.numeric(
29   readline("\nEnter Spend for prediction: ")
30 )
31 new_data <- data.frame(
32   Spend = new_spend
33 )
34 prediction <- predict(
35   model,
36   newdata = new_data
37 )
38 cat("\nPredicted Sales =", prediction)
```

Console Output:

```
> source("C:/Users/2021e/Desktop/EX 1.R")
Enter number of records: 5
Enter Spend 1 : 100
Enter Sales 1 : 10
Enter Spend 2 : 200
Enter Sales 2 : 20
Enter Spend 3 : 300
Enter Sales 3 : 30
Enter Spend 4 : 400
Enter Sales 4 : 40
Enter Spend 5 : 500
Enter Sales 5 : 50

Advertising Data:
  Spend Sales
1  100    10
2  200    20
3  300    30
4  400    40
5  500    50

Regression Model:

Call:
lm(formula = Sales ~ Spend, data = advertising)

Coefficients:
(Intercept)          Spend 
 1.271e-14    1.000e-01 

Enter Spend for prediction: 2
Predicted Sales = 0.2
```

Environment Pane:

Object	Class	Attributes
advertising	data.frame	5 obs. of 2 variables
model	lm	List of 12
new_data	data.frame	1 obs. of 1 variable
new_spend	numeric	1 obs. of 1 variable